

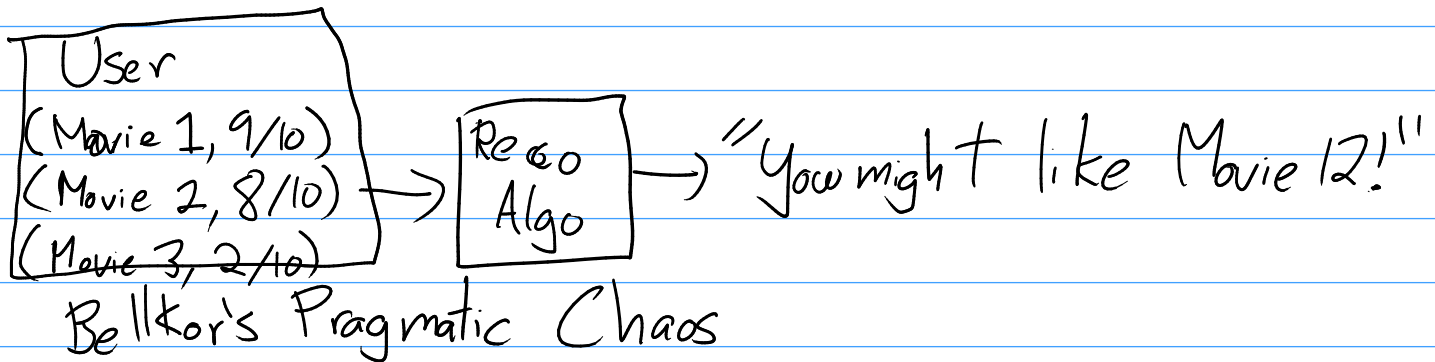
A Course on Differential Privacy
Gautam Kamath, University of Waterloo
@theGautamKamath
www.gautamkamath.com

NYC Taxicab Data

- NYC Taxi and Limo Commission
- Freedom of Info
- Chris Whong
- Jason Hall
 - ↳ very prolific driver
- Pandurangam
 - medal #s and licenses are md5 hashes
 - try all strings!
 - XL589
 - ≈ 27 million possible #'s
- Cross-reference attack
 - match medal # to name
- What if: random ids
 - Take a ride w/ Bob, on 9/28 at 5:23 PM
 - Cross ref w/ data $\rightarrow$ Bob 4
- Coworker leaves at 5:03 PM, on 9/29, left from work
- lookup destination

2. Netflix Prize

- Recommendation engine
- \$1 mill if you make it better



Bellkor's Pragmatic Chaos

(User id, movie, rating, date)

- Narayanan + Shmatikov 2008
attack

'Anonymized' dataset

Netflix data

↑
deidentified

Public datasets for x-ref

IMDb

↑
names, pseudonym

- Cancellation of sequel Netflix Challenge
- Class action lawsuit

3. Memorization in Neural Network

Carlini et al. 2019

Train language model

- dataset $\mathcal{Y}$ (maybe sensitive)
- Want to output f_θ , a generative seq. model

Given $x_1, \dots, x_n$, f_θ can compute

$$P_\theta(x_1, \dots, x_n) = -\log_2 P(x_1, \dots, x_n | f_\theta) = \sum_{i=1}^n (-\log_2 P(x_i | f_\theta(x_1, \dots, x_{i-1})))$$

↑
log-perplexity

Low perplexity $\leftrightarrow$ high prob. by f_θ

(High perplexity $\leftrightarrow$ less likely according to f_θ)

→ "Mary had a little lamb" (if f_θ is a good model of eng.)
→ "Correct horse battery stapler" (not likely sentence in English)

↑
If it were in training data?

Seriously "my social security number is 078-05-1120"

1. Do NNs "memorize" "secrets" in training data?
2. Can these secrets be found efficiently?

Experiment:

- Canary:
- Is the canary lower perp. than
"my social security number is 867-53-0900"

Set of phrases R

$R = \{ \text{"my social security number is ???-??-????"} \}$

- Sort all of R , in increasing log-perplexity
- The rank of the canary (wrt θ) is its ~~ex~~ index in this list
- "exposure"
of r $= \log_2 |R| - \log_2 \text{rank}(r)$.
- Ranges from 0 to $\log_2 |R|$
- Efficient ways of finding exposed strings
- How to mitigate?
 - Dropout, regular X
 - Differential Privacy ✓

4. Genomic or Medical Studies

- Very important, very sensitive
- Test/Study for HIV positive indiv.
- Smokers $\rightarrow$ insurance premium $\uparrow$
- Homer et al. 2008
 - $\hookrightarrow$ resolve individuals in certain studies
- $\rightarrow$ As a result, NIH stopped publishing
 - MAF, χ^2 -test, p-values $\times$
- $\rightarrow$ ~~Reveal~~ Privacy-preserving analyses $\rightarrow$ open Science

5. Mass GIC

- Free data sharing of hospital visits
- Bill Weld: Don't worry!
- Anonymized...

- Orig dataset → (name, ~~SSN~~, ZIP, dob, sex, condition)

- Anon:

- Latanya Sweeney

- \$20 → bought voter rolls

(name, address, ZIP, dob, sex)

- Reidentified Weld

Uniquely identifies 87% of Americans

k-Anonymity [Samarati - Sweeney '98]

(~~name~~, zip, nationality, age, condition)

↑ identifier ↑ sensitive

 pseudo-identifiers 4-anonymous 6-anonymous

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<30	*	AIDS
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
5	130**	≥40	*	Cancer
6	130**	≥40	*	Heart Disease
7	130**	≥40	*	Viral Infection
8	130**	≥40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

	Non-Sensitive			Sensitive
	Zip code	Age	Nationality	Condition
1	130**	<35	*	AIDS
2	130**	<35	*	Tuberculosis
3	130**	<35	*	Flu
4	130**	<35	*	Tuberculosis
5	130**	<35	*	Cancer
6	130**	<35	*	Cancer
7	130**	≥35	*	Cancer
8	130**	≥35	*	Cancer
9	130**	≥35	*	Cancer
10	130**	≥35	*	Tuberculosis
11	130**	≥35	*	Viral Infection
12	130**	≥35	*	Viral Infection

Def: A dataset is k -anonymous if, for any setting of the pseudo-identifiers, there are either 0 or $\geq k$ rows with that setting.

- Anonymization doesn't work
(ad hoc)
- k-anonymization - side information are bad
- even when we learn a model of f_n of data,
still leak private info